

DIG320 Simulation Design

Semester: Fall 2026

Location: Roessner 200

Days/Times: M/W/F 1:00-1:50am

Instructor: Mark Baldridge

Email: mbaldridge@albright.edu

Student Hours: Fridays 9am to Noon —other times by appointment

Office: 345 Science Hall

PRO-TIP: *When emailing any professor about a course, begin the "Subject" line with the course number or name -then clearly state your reason for writing*

Introduction

This semester Simulation will focus on a particular space on campus: *Gingrich Library*, and *Joyce Family Foundation Learning Commons*

This will give a real-world flavor to our designs and simulations. We'll also try out some new technology: Besides modeling, mapping, texturing and lighting, we'll look into newer forms of modeling, including *gaussian splat modeling* which resolves very many photographs as an interactive 3D model. We may also look at 3D tools like Meshy.

Modeling and simulation are at an exciting time and this class wants to sample all the pipelines leading to great 3D simulations of real or imagined spaces.

As such we'll break down the course in to three categories, tho we may approach each of them in a non-linear manner

- **Traditional Digital Pipeline** Modeling in Maya 3D, adding various maps including "normal maps" or "height" maps which add illusions of depth, texture maps which tend to be raster graphic "pictures" of textures and lighting effects to create the sense of volume, space, mass and "place"
- **Gaussian Splat** An advanced 3D reconstruction and rendering technique that turns a collection of ordinary photos or video frames into an

interactive, photorealistic 3D scene. Instead of using hard polygon meshes, it represents space using millions of soft, semi-transparent 3D ellipsoids ("splats").

- **AI or Other Pipelines** Creating high-quality 3D assets has traditionally been one of the most difficult and time-consuming tasks in game development, AR/VR projects, animation, product design, and 3D printing. Artists often spend hours or even days building models, creating textures, rigging characters, and optimizing assets for production use. AI tools want to change that. Are they good enough?

Goals and Objectives

Instructional Goals:

- Build a foundation in the fundamentals of simulating surfaces and spaces
- Push the boundaries of the new technology to create convincing representations of real places
- Interrogate the possibilities of AI methods for generating practical environments from images and prompts

Learning Objectives

- **Technical Proficiency** Learnt the hands-on techniques involved in 3D modeling and representation
- **Conceptual** Explore the boundaries of the technology and techniques for making immersive spaces
- **Developmental** Explore future technologies and techniques, identify their powers and limitations and speculate about their future development

Computing Requirements

If you own a gaming laptop, you should bring it to class. The classroom should come with Photoshop, Maya, Unity and other software installed but your own compute will be best managed by you.

Organization

Portfolio

You will never email or give me your assignment (unless asked, for troubleshooting purposes) but will keep and preserve your work in a digital portfolio.

For source files, you may use a google drive — for video files, a youtube channel. Tho if you prefer ftp and Vimeo or some other combo, this is up to you!

I don't wish you to preserve your work in any space tied to your student email. *I want you to have access to this material long after graduation, and indeed until you decide to archive or delete it!*

If a Google Drive folder, your instructor will want it to be shared by a particular, preferred method:

LINK: [How to share a Google Drive folder with your instructor](#)

You are required to follow this tutorial if using Google Drive!

This kind of digital portfolio is only a few steps away from a *professional* version, useful for applying for gigs, gradschool or etc.

Your professor will always be eager and ready to help you develop a professional portfolio!

Projects

In this course we will produce three Projects, here's a guess at what they might look like

- **Hand Modeled Space** A 3d representation of the chosen space, represented more or less realistically (that is actually a choice left up to you how realistic you want this to be)
- **Compiled Space** Using new techniques to compile many photographs into a realistic representation of an existing space
- **Advance!** Exploring advanced, likely lose or difficult to predict tools to try to make something useable in virtual spaces

Evaluation

All course work will be evaluated on the basis of a 100-point semester

- **Tutorial Assignments**
 - Each of these is calculated to lead to a particular Project
 - The number of these may vary, but for each of the three sections, assignments will always total 10 points (if there are five Tutorial Assignments leading up to *Abstractions*, each will be worth 2 points, etc)
- **Three Projects**
 - As described and named above
 - Each worth a maximum of 20 points
- **The remaining 10 points are divided into:**
 - *Creativity Points* (3)
 - Show design-centric work which engages viewers on an aesthetic level
 - *Innovation Points* (3)
 - Demonstrate techniques or approaches not covered in class, seek to educate yourself beyond the coursework and demonstrate that knowledge
 - *Engagement Points* (4)
 - Show up to and engage in classroom conversations, ask and answer questions, *help your peers* (this last is always the most impressive to your instructor)

If, early on, you have only one Creativity point, for example, be aware that this grade will not be complete until the semester ends!

Details of Scale

My evaluation scale looks like this:

A grades	B grades	C grades
A+ : 98-100	B+ : 88-90	C+ : 78-80
A : 94-97	B : 84-87	C : 74-77
A- : 91-93	B- : 81-83	C- : 71-73

As this may not be a scale you've seen before, it's worth noting that:

The range of A- begins at 91 points

B- begins at 81

C- at 71

There are no gradations of the D eval, but the range is 60-70

I'm Happy to discuss my grading system, or any particular grade!

Tracking Grades

Grades will be posed to after each Project comes to completion but it's the grades (midterm and final) posted on Self Service which are considered the student's official grades

Calendar Fall 2026

WEEK	GOAL	NOTE
1. 08/24		Friday 28, last day to add or change a course
2. 08/31		
3. 09/07		
4. 09/14		Friday 18, Quality/Non election ends
5. 09/21		
6. 09/28		
7. 10/05	Project One Due	MIDTERMS
8. 10/12		FALL BREAK (Monday, Tuesday)
9. 10/19		GAME JAM opens Sunday / Wednesday, Midterm grades due
10. 10/26		GAME JAM due at 12pm Monday 26, Awards tuesday at 7pm, Tav Shande Judges / Spring registration begins
11. 11/02		
12. 11/09	Project Two Due	
13. 11/16		Monday 16, Last day to withdraw with W

WEEK	GOAL	NOTE
14. 11/23		Wednesday 25 THANKSGIVING BREAK
15. 11/30		Friday 4, LAST CLASS
16. 12/07	Project Three Due	FINALS WEEK

Later Dates

Last Incomplete request: Dec 14 FINAL GRADES DUE: Dec 16 Last Q/Non@
 Reversal: Jan 8 Last day to resolve "Incomplete": Feb. 9
